

# Marketing ROI & Budget Reallocation (Attribution + Regression)

## Background

You are a **Business Analytics Associate** at an e-commerce company. The company spends money across multiple marketing channels (Search, Paid Social, Email, Referral, Organic) using multiple campaigns.

Leadership wants a **decision-grade answer**:

1. **Which channels/campaigns are actually driving revenue and orders?**
2. **How should we reallocate next month's budget to maximize revenue (or profit proxy)?**
3. **How confident are we in the recommendation (assumptions + sensitivity)?**

You will receive raw, messy data (spend + sessions + orders + campaigns) and must build a clean analytics layer, do attribution and regression analysis, estimate impact, and present a final budget plan.

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## Datasets Provided

You will receive a folder containing:

1. `users.csv` — user profiles & signup info
2. `campaigns.csv` — campaign metadata (channel, objective, start/end dates)
3. `ad_spend_daily.csv` — daily spend + performance (impressions, clicks) by campaign
4. `sessions.csv` — session metadata (channel/campaign/device)
5. `orders.csv` — purchases (only for converted sessions)
6. `order_items.csv` — basket composition
7. `products.json` — product catalog (category, price, cost)

Note: Data contains realistic issues (missing values, duplicates, inconsistent casing, outliers). Cleaning is part of the project.



Download the dataset from here:

<https://drive.google.com/drive/folders/1R4y5SqFLIByLOaNinarPHxURZxvrqXMy?usp=sharing>

## Your Task (End-to-End)

### Part A — Problem Framing

Create a framing document that includes:

- **Business Objective** (maximize revenue or margin-proxy under a fixed budget)
- **North Star Metric** (choose one and justify), examples:
  - Incremental revenue (proxy) per ₹ spent
  - Contribution margin (proxy) per ₹ spent
- **Supporting KPIs (8–12)**, must include:
  - ROAS (revenue/spend)
  - CAC proxy (spend/orders or spend/new customers)
  - Conversion rate (session → order)
  - Revenue per session
  - AOV
  - Discount rate
  - Margin proxy (if using cost from products)
  - Channel mix share (spend and revenue)
- **Attribution rule** you'll use (e.g., last-touch using session channel/campaign)
- **Stakeholder questions** (5–7) you will answer with data

### Part B — Data Architecture & ETL (Python)

Build a reproducible **Python ETL pipeline** that:

1. Loads all raw files
2. Cleans and standardizes:

- deduplicate sessions/orders/spend rows
- normalize casing for channel fields
- handle missing spend/clicks/impressions
- flag revenue outliers and handle them consistently

1. Produces **three curated datasets**:

### Output 1: **fact\_sessions.csv** (1 row per session)

Must include:

- session\_id, user\_id, session\_ts, device, channel, campaign\_id
- is\_new\_user (based on signup\_date)
- purchase flag + order\_id (if any)
- revenue fields (0 if no purchase): gross, discount, net
- optional: session-to-order time (if applicable)

### Output 2: **fact\_campaign\_daily.csv** (1 row per date per campaign)

Must include:

- date, campaign\_id, channel
- spend, impressions, clicks
- attributed sessions, attributed orders, attributed revenue (from your attribution logic)
- derived KPIs: CPC, CTR, CVR, ROAS, CAC proxy

### Output 3: **fact\_channel\_daily.csv** (1 row per date per channel)

Must include:

- date, channel
- total spend, attributed orders, attributed revenue
- control variables:
  - day\_of\_week
  - promo\_flag (derive from data trends OR use a simple rule you define and document)
  - seasonality/trend features (e.g., week index)

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## Part C — Analytics (Insights + KPIs)

## 1) Marketing Performance Diagnosis

- Overall ROAS / CAC proxy by channel and by campaign
- Identify **top 5 campaigns by spend** and **top 5 by revenue**
- Identify **wasted spend candidates** (high spend + low revenue)

## 2) KPI Trends (Daily/Weekly)

- Weekly trends for **8–12 KPIs**
- Identify anomalies/spikes and propose explanations (seasonality, promos, etc.)

## 3) Segment Deep Dive (Mandatory)

Analyze performance by:

- channel × device
- new vs returning users
- city\_tier / segment (from users)
- top product categories (from order\_items + products)

Deliver:

- top 3 segments where conversion differs significantly
- top 3 segments where ROAS differs significantly
- recommended actions per segment

## 4) Marketing Investigation Table (Mandatory)

Create a table for the **top 2 spend inefficiencies**, containing:

- spend share %, revenue share %, ROAS, CAC proxy
- where the inefficiency is concentrated (channel/campaign/device/segment)
- 2–3 hypotheses (possible causes)
- 1 follow-up experiment or data to validate each hypothesis
- **Evidence used** (metric/chart/table reference)

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## Part D — Attribution Analysis (Must-have)

Implement **Last-Touch Attribution**:

- Attribute each order to the session's channel/campaign (if present)

- Produce:
    - attributed revenue by channel and campaign
    - attributed orders by channel and campaign
  - Compare **attributed ROAS** vs **simple blended ROAS** and explain differences.
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## Part E — Regression Impact Modeling (Must-have)

Build a regression-style model to estimate the relationship between **spend** and **outcomes** while controlling for seasonality.

Minimum requirements:

- Model target: **daily revenue** (or daily orders)
  - Features must include:
    - spend by channel (or campaign aggregates)
    - controls: day-of-week + trend + promo\_flag (or equivalent)
    - optional: lagged spend (1–3 day lag) or log(spend) to reflect diminishing returns
  - Evaluate model fit:
    - train/test split (time-based split preferred)
    - MAE/MAPE (or RMSE) +  $R^2$
  - Interpret results:
    - which channels have highest estimated impact (marginal effect)
    - limitations (correlation vs causation, missing confounders, etc.)
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## Part F — Budget Reallocation Plan + 30-Day Impact Estimate (Mandatory)

Given a fixed next-month budget (you define a budget number and document it), propose:

- a new spend allocation across channels (and/or top campaigns)
- constraints:
  - minimum and maximum spend per channel (you choose reasonable values and justify)
- estimate next-month impact:
  - base/best/worst-case incremental revenue

- margin proxy impact (if using product cost)
  - include assumptions + sensitivity.
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## Part G — Dashboard (Tableau / PowerBI / Excel)

Create a dashboard with **minimum 4 views/pages**:

### View 1: Executive Summary

- North Star + KPI tiles
- channel mix + total spend + total attributed revenue
- next-month budget recommendation + impact estimate

### View 2: Channel & Campaign Performance

- ROAS/CAC by channel
- top campaigns table (spend vs revenue)
- filters/slicers for channel, device, time

### View 3: Attribution vs Regression

- attributed revenue share vs regression-estimated impact (side-by-side)
- explain in 2–3 bullets what the difference means

### View 4: Segments & Opportunities

- new vs returning performance
- device × channel view
- top categories performance by channel

(If using Excel, include pivots + slicers and a clean navigation layout.)

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## Final Narrative (Decision Memo)

Create either:

- **7–10 slide deck**, OR
- **2–3 page memo**

Must include:

1. Goal + KPIs + attribution rule

2. Data + methodology (ETL + definitions)
  3. Key insights (5–7)
  4. Attribution findings (what looks good/bad)
  5. Regression findings (estimated impact + limitations)
  6. Budget plan + 30-day impact estimate (assumptions + scenarios)
  7. Risks/limitations + next steps (experiments/data to collect)
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## Submission Guidelines (What You Must Submit)

Create a folder named:

`<YourName>_Capstone_MarketingROI/`

Include:

### 1) `README.md`

- project overview + north star
- attribution method used
- how to run ETL end-to-end
- curated outputs generated
- dashboard tool used + how to open it

### 2) `/data/` (*generated only*)

- `fact_sessions.csv`
- `fact_campaign_daily.csv`
- `fact_channel_daily.csv`

### 3) `/etl/`

- `etl_pipeline.py` (script preferred)
- helper modules (if any)

### 4) `/analysis/`

- `analysis.ipynb` (or `analysis.py`)
- must include:

- KPI trends
- attribution results
- regression model + evaluation + interpretation
- budget reallocation logic + impact estimate

## 5) `/dashboard/`

One of:

- `.twbx/.twb` OR `.pbix` OR `.xlsx`

Also include:

- `dashboard_screenshots/` (4–6 images) OR one exported PDF

## 6) `/final_story/`

- `final_deck.pdf` OR `final_memo.pdf`

## Google Drive Submission (Important)

- Upload your full submission folder to Google Drive.
  - Share **one public view link** (anyone with the link can view).
  - **Open the link in Incognito/Private mode** to confirm it's accessible without "Request access".
  - Submit that final link.
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